

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

**ENVIRONMENTAL SCIENCE****Paper 2 (Generated practice paper - Set 08)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

**Time allowed: 3 hours****Materials**

For this paper you may use:

- a calculator.

**Instructions**

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

**Information**

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

**For Examiner's Use**

| Question | Mark |
|----------|------|
| 1        |      |
| 2        |      |
| 3        |      |
| 4        |      |
| 5        |      |
| 6        |      |
| 7        |      |
| 8        |      |
| 9        |      |
| 10       |      |
| 11       |      |
| TOTAL    |      |

Answer all questions in the spaces provided.

**0 1 . 1** Explain how each of the following reduces the environmental impact of fishing.  
[8 marks]

**Square-mesh escape panels** .....

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**Minimum and maximum catchable sizes** .....

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**No-take zones** .....

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**Closed seasons** .....

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**0 1 . 2** Explain why catch quotas work well for pelagic species such as mackerel but poorly in mixed demersal fisheries.  
[2 marks]

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**10**

**0 2**

Table 1 shows some of the IUCN criteria used to place species in Red List categories.

**Table 1**

| Category              | Population decline over 10 years or 3 generations | Area of habitat            | Number of mature adults | Probability of extinction in the wild |
|-----------------------|---------------------------------------------------|----------------------------|-------------------------|---------------------------------------|
| Vulnerable            | over 40%                                          | under 2000 km <sup>2</sup> | under 10 000            | over 10% in 100 years                 |
| Endangered            | over 60%                                          | under 500 km <sup>2</sup>  | under 2500              | over 20% in 20 years or 5 generations |
| Critically endangered | over 80%                                          | under 10 km <sup>2</sup>   | under 250               | over 50% in 10 years or 3 generations |

**0 2****1**

State **three** roles of the IUCN.

**[3 marks]**

1 .....

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2 .....

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3 .....

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**0 2****2**

A species of lemur has declined from 12 000 to 2 100 mature adults in three generations and occupies 400 km<sup>2</sup> of forest. Use Table 1 to determine its Red List category. Explain your answer.

**[3 marks]**

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**0 2****3**

Explain why only a small proportion of species have been assessed for the Red List, and give **two** criteria used to decide which species to assess.

**[3 marks]**

0 2 . 4

Give **one** example of a species whose Red List category improved because of conservation and state the actions responsible.

[1 mark]

10

0 3

Table 1 shows some events in the history of life on Earth.

Table 1

| Event                                                 | Time before present / billion years |
|-------------------------------------------------------|-------------------------------------|
| Formation of the Earth                                | 4.6                                 |
| First life (Archaea) on Earth                         | 3.5                                 |
| First photosynthetic organisms                        | 2.7                                 |
| Oxygen concentration in the atmosphere starts to rise | 2.45                                |
| End of the period of oxygen build-up (Proterozoic)    | 0.54                                |

0 3 . 1

Use Table 1 to calculate how many **million** years passed between the appearance of photosynthetic organisms and the start of the rise in atmospheric oxygen.

[1 mark]

Answer

million years

03 . 2

Explain why atmospheric oxygen did not start to rise as soon as photosynthesis began.

[2 marks]

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03 . 3

Explain how the rise in atmospheric oxygen made life on the land surface possible.

[3 marks]

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03 . 4

Explain how early photosynthetic organisms helped to prevent a long-term rise in the Earth's temperature.

[2 marks]

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03 . 5

Give **two** other ways in which living organisms changed the conditions on Earth.

[2 marks]

1 .....

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2 .....

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10

04 . 1

Describe the ecological features of temperate broadleaf woodland and explain how woodland floor plants such as bluebells are adapted to them.

[4 marks]

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04 . 2

Explain why an ancient woodland (existing before 1600) usually has higher biodiversity than a recently planted woodland.

[3 marks]

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04 . 3

Describe **three** threats to UK broadleaf woodland.

[3 marks]

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2 .....

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3 .....

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10

**0 5 . 1** Describe how agriculture alters the hydrological cycle.

**[3 marks]**

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**0 5 . 2** Explain how each of the following consumer choices can reduce the environmental impact of food production.

**[4 marks]**

**Seasonal / local food** .....

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**Organic food** .....

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**0 5 . 3** Explain why post-war European governments introduced grants and guaranteed prices, and describe **one** environmental consequence.

**[3 marks]**

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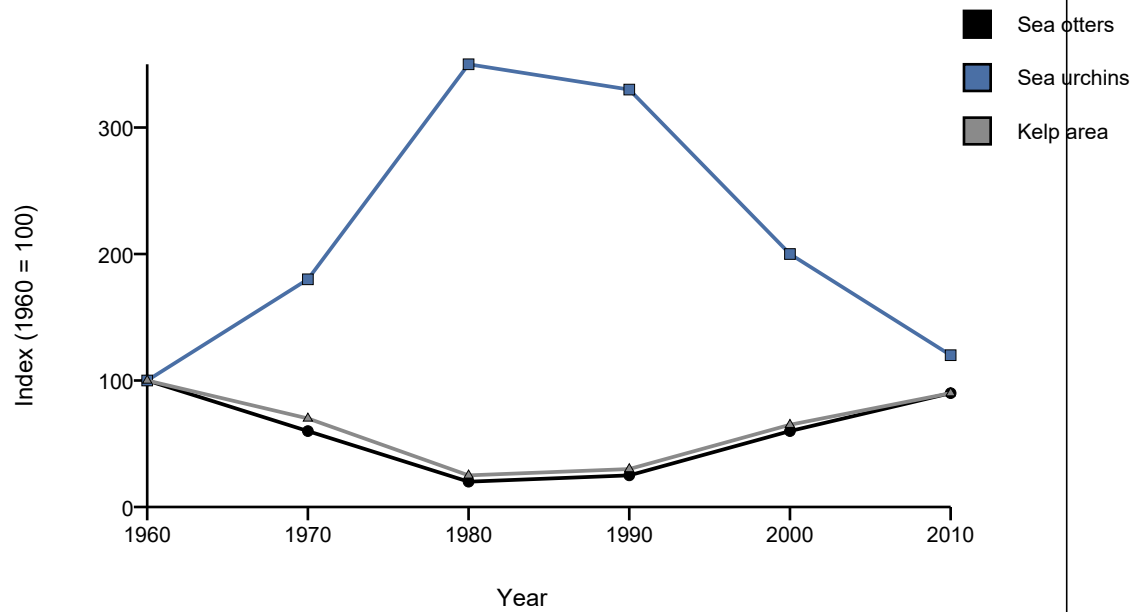
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10

0 6

Figure 1 shows the estimated populations of sea otters, sea urchins and the area of kelp forest along part of the coast of California between 1960 and 2010.

**Figure 1**



0 6

1 Describe the relationship between the three variables shown in Figure 1.

**[3 marks]**

0 6

2 Explain the relationships shown in Figure 1.

**[3 marks]**

06 . 3

Give **two** other examples of human over-exploitation of one species affecting the population of another.

[2 marks]

1

2

06 . 4

Explain why the loss of beavers or forest elephants can cause the decline of many other species.

[2 marks]

10

07 . 1

Complete Table 1 to compare natural systems with typical linear human systems.

[4 marks]

Table 1

| Feature                  | Natural systems | Linear human systems |  |
|--------------------------|-----------------|----------------------|--|
| Energy source            |                 |                      |  |
| Use of materials         |                 |                      |  |
| Nature of wastes         |                 |                      |  |
| Temperature of processes |                 |                      |  |

07 . 2

Explain why natural systems can be sustained using a relatively small number of elements.

[2 marks]

07 . 3

Explain why the human use of minerals makes them difficult to recover and re-use.

[2 marks]

07 . 4

Explain why toxic metals in electronic products cause problems that the wastes of natural systems do not.

[2 marks]

10

08 . 1

Explain the difference between approaches to sustainable lifestyles based on **moral belief** and on **enlightened self-interest**.

[2 marks]

08 . 2

Explain why the overall environmental impact of humans depends on two main factors.

[2 marks]

08 . 3

Give **one** activity that is sustainable with a small population but unsustainable if the population is too large.

[1 mark]

5

09

Table 1 describes three genetically modified (GM) crops.

Table 1

| GM crop            | Gene source                      | Characteristic                           |
|--------------------|----------------------------------|------------------------------------------|
| Bt maize / cotton  | Bacillus thuringiensis bacterium | Produces a toxin that kills insect pests |
| Roundup Ready soya | Agrobacterium                    | Resistant to the herbicide glyphosate    |
| Golden rice        | Daffodil and Erwinia bacterium   | Produces vitamin A in the grain          |

09 . 1

Explain the main advantage of GM over conventional selective breeding.

[2 marks]

09 . 2

For each crop in Table 1, explain one potential benefit.

[3 marks]

**Bt crops**

**Roundup Ready soya**

**Golden rice**

09 . 3

Explain **four** concerns about GM crops.

[4 marks]

09 . 4

Give **one** reason why GM crops have been slow to enter cultivation in Europe.

[1 mark]

10

Evaluate the relative importance of ex-situ and in-situ methods in the conservation of endangered species.

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13

Extra space

10 . 2 State what is meant by **rewilding**.

[1 mark]

10

Do not write outside the box

**Write an essay on one of the following topics.**

|   |   |   |   |
|---|---|---|---|
| 1 | 1 | . | 1 |
|---|---|---|---|

Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them.

**[25 marks]**

**OR**

1 1 . 2

Discuss the extent to which agriculture can be made sustainable while feeding a growing human population.

**[25 marks]**

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.







**END OF QUESTIONS**